

Using Rhythm Game to Train Rhythmic Motion in Sports

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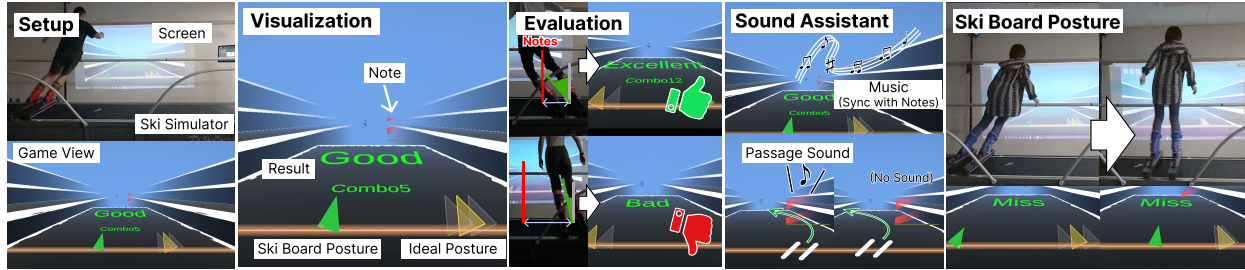


Figure 1: The overview of the rhythm game we developed.

ABSTRACT

Rhythm is important to improve skills in various sports. In this study, we create a rhythm game for ski training. It synchronizes music and turn timing, and has various types of feedback for training. We conducted a pilot study to verify the effectiveness through a comparison of three conditions.

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1 INTRODUCTION

Rhythmic movements are included even in sports not directly related to music. Rhythmic training affects players' performance [Söğüt et al. 2012]. However, most of the current training systems focus on visual feedback of posture differences [Wu et al. 2019].

In this paper, taking alpine skiing as a good example, we study the effect of using rhythm game-like methods to train the motions of the rhythmic sport. The system allows the user to ski in time to the music to maintain a certain interval between turns and provides feedback on the user's posture during the turns to maintain motivation. For comparison, a pilot study was conducted to test three conditions: one with no feedback, one with the sound assistant, and one with a rhythm game.

2 RELATED WORK

For ski training, various feedbacks to support training in the VR ski simulator were discussed [Wu et al. 2019], and a trampoline-based

game was shown to increase user motivation through a variety of effects [Lehtonen et al. 2019]. However, these studies focus on the user's motion or posture error but not on the rhythm of the motion.

There was also a system to evaluate skiers' turns in terms of rhythm [Yamagiwa et al. 2014], and a study that showed that exercise in a certain rhythm positively affected tennis performance [Söğüt et al. 2012]. The goal is to create an effective training system using these effects.

3 GAMIFICATION DESIGN

To test the effectiveness of rhythm game training, we designed a game for alpine skiing played on a ski simulator (Figure 1). It has the following functions.

- (1) **Evaluation:** This system evaluates the user's ski board posture as it passes the gate on four levels: Excellent, Good, Bad, and Miss in terms of its angle, distance from the edge of the gate, and whether the user successfully passed or not.
- (2) **Sound Assistants:** To allow the user to ski in rhythm, the system plays background music, and the timing of the user's passage through the gates is synchronized with the beat of the music. The gates are placed at intervals passing 60 gates per minute using a song with a BPM of 120 to provide the level of difficulty for beginners. In addition, a short sound will be played if the user's posture is rated Excellent or Good.
- (3) **Ski Board Posture:** The current position and angle of the skis are displayed as green markers, and the yellow marker indicates the ideal posture for scoring Excellent. The white markers present the timing to pass the gate.

4 PILOT STUDY

4.1 Study Design

Our target is to verify whether using music and rhythm game can improve ski training. Therefore, the following hypotheses are raised. **H1:** Particular music can improve rhythmic ski performance. **H2:** The rhythm game can improve rhythmic ski performance. **H3:** The rhythm game can motivate the users.

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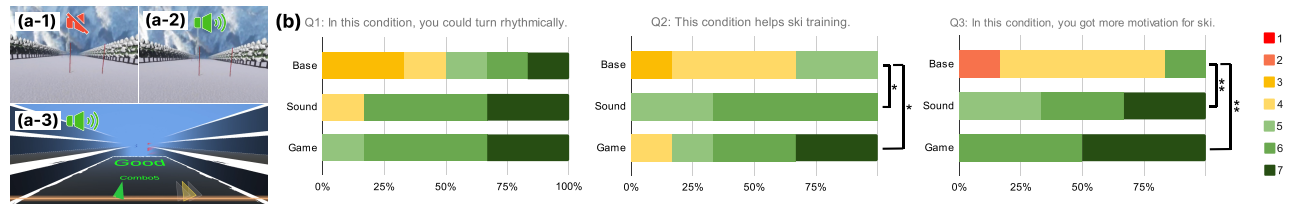


Figure 2: The scenic views in the 3 conditions, *Base* (a-1), *Sound* (a-2), and *Game* (a-3). The results of the three questions in Questionnaire (b). The brackets on the right indicate the significant difference between the conditions: * ($p < 0.05$), ** ($p < 0.01$)

To verify the hypotheses, we designed the following conditions where the visual and audio information are different. **Base** (Figure 2 (a-1)) is a baseline scene with gates on a virtual slope similar to real alpine skiing, without any sound. **Sound** (Figure 2 (a-2)) is a Base scene only with the sound assistants. **Game** (Figure 2 (a-3)) is the rhythm game described in the previous section.

Six participants were invited to the pilot study (three females, three males, mean age=23.5, std=1.98), all with experience in a ski simulator. The procedure of the study was as follows. First, participants were asked to get used to the system in the *Base* condition, and their initial performances were captured. Next, they trained in one condition and then their performances were captured once again in the *Base* condition for comparison. After training, a questionnaire asking about the condition experienced was given to them. This was repeated for all three conditions, while the order was shuffled using the Balanced Latin Square for counter-balance. Finally, an interview about the overall impression was performed.

4.2 Questionnaire Result

The questions (Q1 - Q3) and the results are illustrated in Figure 2 (b). The results of the questionnaire were tested by One-Way ANOVA and found to be significantly different from the results of Q2 and Q3. In Q1, the *Sound* and *Game* conditions tends to get higher evaluation than the *Base* condition. Tukey HSD multiple testing of Q2 and Q3 results indicates that the *Game* and *Sound* conditions are more motivative and better suited for training. Although there was no significance between the *Game* and *Sound* conditions, there was also a slight tendency for the *Game* condition to be more motivative.

4.3 Interview Result

Many participants said that the *Game* condition is motivative. In the *Game* condition, some participants commented that the ski board posture is good to understand when to turn, and other participants often focused on the score visualization. Some participants also claimed that some feedback such as passage sound and ideal posture was out of sync with the actual timing of the turns.

5 DISCUSSION

The pilot study results and user feedback provided insight into the aforementioned hypotheses H1-H3.

H1: The results of the Q1 questionnaire show that the method using music tends to improve skiing performance, although there is no significant difference. In particular, some participants were listening carefully to the music when they first started sliding, suggesting that music is effective in grasping turn timing.

H2: The results of Q2 show that the *Game* condition has the same training effects as the *Sound* condition. If the issues mentioned in the interview, such as the discrepancy between the timing of turns, could be improved, the practice effect could be even better.

H3: The results of Q3 and the interviews also indicate that the *Game* condition is significantly more motivating than the *Base* condition, and furthermore, the *Game* condition tends to be rated better than the *Sound* condition. One reason may be that the goal is made clearer by the display of the ski board posture and the score.

Since the study was conducted with a small number of participants within-subjects, only a rough trend could be obtained. In the future, we would like to conduct a larger group between-subjects study to investigate the effects of learning rhythmic sense in more detail. We also plan to incorporate more skiing motions into the game and add other postures such as foot pressure and upper body to the evaluation to make the game more effective for training.

6 CONCLUSION

This study proposed a method using a rhythm game as a training program for fostering a sense of rhythm in an alpine ski. The pilot study showed that a ski training system using a rhythm game can provide similar training effects despite the fact that the screen is completely different from actual skiing, indicating that there is room for more detailed research on this method.

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